ESP32-S3 Series Datasheet - DATASHEET

Manufacturer: Espressif
Part Number: ESP32-S3
Category: Microcontroller / WiFi+BLE MCU
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Overview

ESP32-S3 is a dual-core XTensa LX7 microcontroller with integrated Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE) connectivity. CPU clock frequency up to 240 MHz. Features include a vector instruction extension for accelerating neural network computing and signal processing. Internal memory: 512 KB SRAM. External memory support: up to 1 GB Flash and 32 MB PSRAM via SPI/OSPI.

Peripheral Interfaces

GPIO: 45 programmable GPIOs. SPI: 4× SPI interfaces (SPI0, SPI1, SPI2, SPI3). I2C: 2× I2C interfaces. UART: 3× UART interfaces. I2S: 2× I2S interfaces. LED PWM: 8 channels. MCPWM: 2 modules. USB OTG: 1× Full-Speed USB 2.0. ADC: 2× 12-bit SAR ADCs, up to 20 channels. DAC: none (removed from S3). Touch sensor: 14 capacitive touch GPIOs.

Wi-Fi Specifications

Wi-Fi Standards: 802.11 b/g/n (2.4 GHz). Data rate: up to 150 Mbps. TX power: 21 dBm. RX sensitivity: -97 dBm (11b, 1 Mbps). Security: WPA3-Personal, WPA3-Enterprise, WPA2-Personal, WPA2-Enterprise, WEP. Protocols: IPv4, IPv6, SSL/TLS, TCP, UDP. Wi-Fi modes: Station, SoftAP, Station+SoftAP. Wi-Fi Direct (P2P) supported.

Key Specifications Summary

cores: 2xLX7. freq: 240MHz. sram: 512KB. wifi: 802.11bgn. ble: 5.0. gpio: 45. adc: 20ch_12bit. usb: otg.